

OS LAB-7
DATE: 27/4/26

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BATCH: 4K2

1) Producer-Consumer Problem using Semaphores

```
#include <stdio.h>
#include <stdlib.h>
#include <pthread.h>
#include <semaphore.h>
#include <unistd.h>

#define SIZE 5

int buffer[SIZE];
int in = 0, out = 0;

sem_t empty;
sem_t full;
pthread_mutex_t lock;

void* producer(void* arg) {
    for (int i = 0; i < 5; i++) {
        int item = rand() % 100;

        sem_wait(&empty);
        pthread_mutex_lock(&lock);

        buffer[in] = item;
        printf("Produced %d\n", item);
        in = (in + 1) % SIZE;

        pthread_mutex_unlock(&lock);
        sem_post(&full);

        sleep(1);
    }
    return NULL;
}

void* consumer(void* arg) {
    for (int i = 0; i < 5; i++) {
        sem_wait(&full);
        pthread_mutex_lock(&lock);
```

```

    int item = buffer[out];
    printf("Consumed %d\n", item);
    out = (out + 1) % SIZE;

    pthread_mutex_unlock(&lock);
    sem_post(&empty);

    sleep(2);
}
return NULL;
}

int main() {
    pthread_t prod, cons;

    sem_init(&empty, 0, SIZE);
    sem_init(&full, 0, 0);
    pthread_mutex_init(&lock, NULL);

    pthread_create(&prod, NULL, producer, NULL);
    pthread_create(&cons, NULL, consumer, NULL);

    pthread_join(prod, NULL);
    pthread_join(cons, NULL);

    sem_destroy(&empty);
    sem_destroy(&full);
    pthread_mutex_destroy(&lock);

    return 0;
}

```

Produced 41

Consumed 41

Produced 67

Produced 34

Consumed 67

Produced 0

Consumed 34

Produced 69

Consumed 0

Consumed 69

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Producer - Consumer using Semaphores

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```
#include <stdio.h>
#include <stdlib.h>
#include <pthread.h>
#include <semaphore.h>
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#define SIZE 5
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```
int buffer[SIZE];
int in = 0, out = 0;
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sem_t empty;
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void* producer(void* arg) {
    for(int i = 0; i < 5; i++) {
        int item = rand() % 100;
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```
        sem_wait(&empty);
        pthread_mutex_lock(&lock);
```

```
        buffer[in] = item;
        printf("Produced %d\n", item);
        in = (in + 1) % SIZE;
```

```
        pthread_mutex_unlock(&lock);
        sem_post(&full);
```

```
        sleep(1);
    }
```

```
    return NULL;
}
```

void*
for

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```

void* consumer(void* arg) {
    for (int i = 0; i < 5; i++) {
        sem_wait(&full);
        pthread_mutex_lock(&lock);
        int item = buffer[out];
        printf("Consumed %d\n", item);
        out = (out + 1) % SIZE;
        pthread_mutex_unlock(&lock);
        sem_post(&empty);
        sleep(2);
    }
    return NULL;
}

```

```

int main() {
    pthread_t prod, cons;

    sem_init(&empty, 0, SIZE);
    sem_init(&full, 0, 0);
    pthread_mutex_init(&lock, NULL);

    pthread_create(&prod, NULL, producer, NULL);
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    pthread_join(prod, NULL);
    pthread_join(cons, NULL);

    sem_destroy(&empty);
    sem_destroy(&full);
    pthread_mutex_destroy(&lock);
    return 0;
}

```


Output: (10 * 6) = 60
 2 * 10 = 20 = 100

Produced (41) from - mod

Consumed 41 return - beauty

Produced 67

Consumed 67 = not a tri

Produced 34 beauty")

Produced 0 (1 + 2) = two

Consumed 34

Produced 69 return beauty

consumed 0 (2) tag - mod

Consumed 69

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